

Rohan Surana

San Diego, CA | 510-265-4239 | rohansurana2810@gmail.com | [linkedin.com/in/rohansurana28/](https://www.linkedin.com/in/rohansurana28/) | <https://rohan2810.github.io/>

Summary

Researcher at Qualcomm working on proactive, personalized LLM agents and long-horizon context management and memory. Published research on RL post-training, preference optimization, and agent evaluation at ICML, COLM, ICLR, and ACL.

Education

Masters of Science in Data Science

University of California, San Diego (UCSD) GPA: 3.88/4.00. Advisor: Julian McAuley

March 2026

San Diego, CA

Bachelor of Science in Software Engineering

San Jose State University, GPA: 3.87/4.00 (Summa cum laude)

May 2022

San Jose, CA

Experience

Qualcomm AI Research

Research Scientist

San Diego, CA

Apr 2026 – Present

- Research on proactive agents and long-range context management to improve agentic capabilities

Dell Technologies

AI Research Intern

Hopkinton, MA

June 2025 – Sept 2025

- Built multi-agent LLM system (LangGraph/LangChain + vLLM) with request batching and KV-cache, cutting p95 latency by 40%
- Designed scalable RAG pipelines integrating LLMs with vector databases using hybrid search and intelligent chunking strategies
- Engineered short/long-term memory with retrieval caching to reduce serving cost & exposed APIs for integration with existing workflows
- Developed agent-based monitoring using MCP and A2A protocols to automate telemetry collection, anomaly detection, and remediation

Dell Technologies

Software Engineer II

Santa Clara, CA

Mar 2024 – Aug 2024

- Architected TOSCA-based framework to orchestrate resources, cutting provisioning time 30% and improving infrastructure flexibility
- Built real-time infrastructure digital twin with predictive analytics using graph databases to enable drift detection and safe remediation

Software Engineer I

Jul 2022 – Mar 2024

- Accelerated cluster time-to-ready by 20% through automated Kubernetes operators supporting GPU workloads, KServe, and TorchServe
- Enhanced service performance by redesigning API and proto services with gRPC, leading a 4-intern team to boost efficiency 15%

Selected Publications

- WS-GRPO: Weakly-Supervised Group-Relative Policy Optimization for Rollout-Efficient Reasoning**
G. Mundada*, Z. Huang*, Rohan Surana*, S. Yu, J. Y. Zhang, X. Li, T. Yu, L. Yao, J. Shang, J. McAuley, J. Wu — *ICML 2026*
- Evaluating Language Model Pluralism through In-the-wild Crowd Discussions**
G. Mundada, Rohan Surana, N. Swaminathan, B. P. Majumder, J. Wu, J. McAuley, Z. Xie — *ACL 2026*
- Importance Sampling for Multi-Negative Multimodal Direct Preference Optimization**
X. Li, C. Wang, J. Wu, Rohan Surana, T. Yu, J. McAuley, J. Shang — *ICLR 2026*
- In-context Ranking Preference Optimization**
J. Wu*, Rohan Surana*, Z. Xie, Y. Shen, Y. Xia, T. Yu, R. Rossi, P. Ammanabrolu, J. McAuley — *COLM 2025*
- Traceable and Explainable Multimodal Large Language Models: An Information-Theoretic View**
Z. Huang, J. Wu, Rohan Surana, R. Jain, T. Yu, R. Addanki, D. Arbour, S. Kim, J. McAuley — *COLM 2025*
- Image Difference Captioning via Adversarial Preference Optimization**
Z. Huang, J. Wu, Rohan Surana, T. Yu, D. Arbour, R. Sinha, J. McAuley — *EMNLP 2025*
- MASS-DPO: Multi-Negative Active Sample Selection for Direct Policy Optimization**
Rohan Surana*, X. Li, S. Yu, Y. J. Shen, C. Wang, T. Yu, P. Ammanabrolu, J. Shang, J. McAuley, J. Wu — *Under Review*
- F-GRPO: Factorized Group-Relative Policy Optimization for Unified Candidate Generation and Ranking**
Rohan Surana, G. Mundada, J. Wu, X. Li, Y. Jiao, B. Jin, S. Zhou, T. Yu, R. Sinha, J. Han, J. Shang, J. McAuley — *Under Review*
- Skill-R1: Agent Skill Evolution via Reinforcement Learning**
Y. Vishe, Rohan Surana, X. Jiang, Z. Huang, X. Li, N. L. Kuang, T. Yu, R. A. Rossi, J. Shang, J. McAuley, J. Wu — *Under Review*
- Skill-CMIB: Multimodal Agent Skill for Consistent Action via Conditional Multimodal Information Bottleneck**
Z. Huang, J. Wu, T. Yu, Q. Yan, Rohan Surana, U. Bhattacharya, L. Yao, X. E. Wang, J. McAuley — *Under Review*
- MusiCRS: Benchmarking Audio-Centric Conversational Recommendation**
Rohan Surana*, A. Namburi*, G. Mundada*, A. Lal*, Z. Novack, J. McAuley, J. Wu — *ICASSP 2026*
- An Empirical Study on Zero-Data Bootstrapping for Conversational Recommender Systems**
Rohan Surana*, J. Wu*, Z. Xie*, Y. Xia, N. Kallus, J. McAuley — *EMNLP 2026 (Findings)*

Technical Skills

- Languages:** Python, Golang, Java, Scala, SQL
- ML/DL:** PyTorch, TensorFlow, scikit-learn, Transformers, OpenCV, NumPy, Pandas
- LLM/NLP:** LangGraph, LangChain, vLLM, Unsloth, CrewAI, Hugging Face
- Infrastructure/Frameworks:** Kubernetes, Docker, FastAPI, Apache Spark, gRPC, Git, MLflow, Ray, AWS, GCP, CI/CD, Postman
- Databases:** ChromaDB, MySQL, PostgreSQL, DGraph, Neo4j, MongoDB